

Exploring Customer Segments in Credit Card Data

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Abstract - This paper evaluates credit card customer segmentation using unsupervised learning and dimensionality reduction techniques on a dataset of approximately 9,000 active cardholders. Since the data lacks predefined labels, K-Means clustering ($k=4$) was first applied to establish baseline customer segments. The methodology presents a detailed pipeline: outlier detection with IQR-based winsorization, Pearson correlation analysis, z-score normalization, and feature selection using Fisher Distance. Then, both PCA and LDA were implemented; evaluation via Silhouette Score and Fisher's criterion demonstrates that LDA yields superior inter-class separation by leveraging the generated cluster labels. Furthermore, we compared K-Means and DBSCAN on the PCA-reduced space against t-SNE and Self-Organizing Maps (SOM) on the original feature space. Results indicate that DBSCAN struggles due to the absence of distinct density gaps, while a two-sample t-test on the credit limit feature confirms statistically significant differences between segments.

Keywords - customer segmentation, K-Means, PCA, LDA, DBSCAN, t-SNE, Fisher Distance, credit card dataset

I. INTRODUCTION

Customer segmentation is crucial for corporate strategies to distinct behavioral patterns. However, real-world transactional data rarely contains predefined categories, rendering method selection a critical determinant of the resulting segments. This study utilizes the Credit Card Dataset for Clustering dataset^[1], capturing 17 behavioral features - including outstanding balance, purchase frequency, cash advance usage, and credit limit - from approximately 9,000 active credit card holders over a six-month period. To address the lack of ground-truth labels, K-Means clustering ($k=4$) was initially applied, guided by the Elbow method and Silhouette analysis. These algorithmic assignments subsequently served as class labels for the downstream analyses. While this introduces a degree of circularity, it remains a pragmatic and accepted paradigm for exploratory research on unlabeled datasets.

Following label generation, an exploratory data analysis applied. Data cleaning incorporated boxplot visualization, Interquartile Range (IQR)-based outlier winsorization, and Pearson correlation analysis to identify feature redundancies. The feature space was then normalized via z-score standardization, and the discriminative power of each feature was quantified using Fisher Distance.

The core of this research evaluates the interplay between dimensionality reduction and class separability. To apply all the requirements described on the assignmnet's paper, Principal Component Analysis was applied to the normalized

space to investigate whether eigenvalue magnitude correlates with discriminative power; consistent with theory, the results confirm that PCA maximizes global variance rather than class separation. Conversely, Linear Discriminant Analysis was employed to project the data into a lower-dimensional space optimized for class distinction. The geometric structures captured by PCA and LDA were rigorously compared using scatter plots, Silhouette Scores, and Fisher's criterion.

For the clustering benchmark, K-Means and DBSCAN were deployed on the two-dimensional PCA projection, while t-SNE and SOM were evaluated on the original 17-dimensional space. The significant failure of DBSCAN to find clear density gaps inherently reveals that credit card behavioral data forms a continuous distribution rather than isolated density pockets. Finally, an independent two-sample t-test was conducted on the CREDIT_LIMIT feature across two distinct segments, confirming statistically significant behavioral variances.

II. METHODOLOGY

A. DATASET DESCRIPTION AND PREPROCESSING

The study utilizes the "CC General dataset" from Kaggle^[1], comprising behavioral data from 8,950 active credit card holders captured over a six-month period. The raw dataset contains 18 attributes: a categorical identifier (CUST_ID) and 17 numerical behavioral features reflecting spending, purchasing, and payment patterns displayed below:

BALANCE	INSTALLMENTS PURCHASES	PURCHASES TRX
CASH ADVANCE	BALANCE FREQUENCY	CREDIT LIMIT
MINIMUM PAYMENTS	PURCHASES FREQUENCY	PAYMENTS
ONEOFF PURCHASES	ONEOFF PURCHASES FREQUENCY	PURCHASES
CASH ADVANCE TRX	CASH ADVANCE FREQUENCY	TENURE
PRC FULL PAYMENT		

The CUST_ID column was excluded from subsequent analyses as it carries no statistical relevance. Initial data screening identified missing values in two attributes: MINIMUM_PAYMENTS lacked 313 entries (~3.5% of the total rows) and CREDIT_LIMIT missed a single value. To preserve sample size, missing values were imputed using feature-wise medians. The median was selected over the mean due to its robust resistance to the heavy right-skewness exhibited by several distributions in the dataset. Code blocks that shows the related column details displayed in the Figure I.

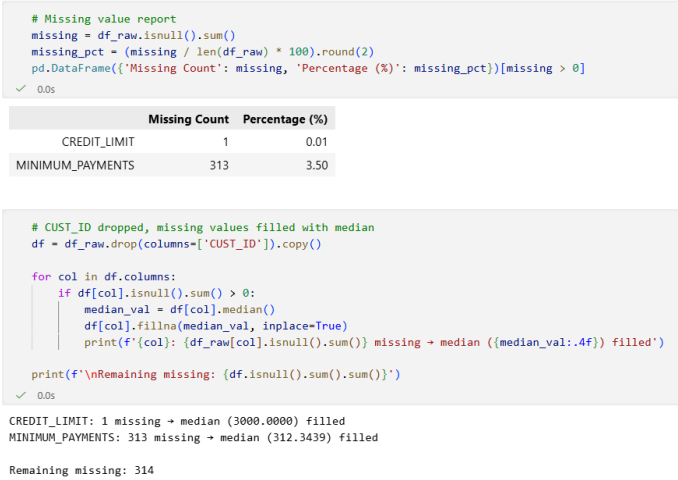


FIGURE I: Dataset Details

Consequently, K-Means clustering was executed on the standardized 17-dimensional feature space. The optimal number of clusters was determined by evaluating the Elbow method alongside the Silhouette Score. Both heuristics converged at k=4, yielding a robust baseline partitioning. These cluster assignments were appended to the dataset as a new discrete target variable (CLASS) and treated as ground-truth labels for the remainder of the analysis. Optimal k selection visualization is displayed in the Figure II.

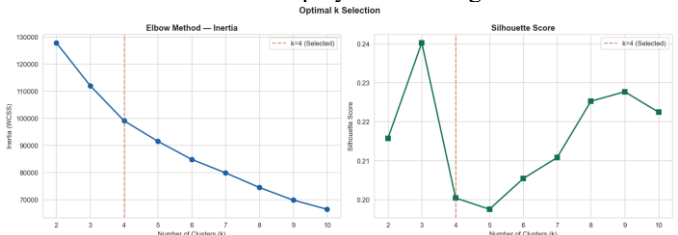


FIGURE II: Optimal k – number of clusters selection

B. Feature Distribution Analysis and Outlier Treatment

To analyze feature distributions across the four generated segments, boxplots for all 17 attributes were constructed on a grid. Each plot was stratified by cluster assignment to visually assess differences in center and variance, while identifying potential extreme values. Monetary attributes - specifically BALANCE, PURCHASES, CASH_ADVANCE, PAYMENTS, and MINIMUM_PAYMENTS- exhibited dense concentrations of data points beyond the upper whiskers.

Outlier detection was formalized utilizing the Interquartile Range (IQR) method, where anomalies were defined as points falling beyond 1.5 times the IQR from the first and third quartiles. The proportion of outliers per feature was subsequently quantified via a horizontal bar chart. To mitigate these anomalies without introducing severe sample attrition through row deletion, a winsorization strategy was deployed. Extreme values were symmetrically clipped to the upper and lower IQR thresholds, preserving the total sample size while constraining the leverage of variance-inflating observations. Before-and-after boxplot comparisons for the six most distorted attributes verified that post-winsorized distributions achieved compact variance while maintaining their fundamental behavioral geometry. The processed dataset was exported as cc_cleaned.csv for later usages. Feature distribution and outlier report are detailly visualized in the jupyter notebook.

C. Correlation Analysis

To evaluate structural dependencies within the data, both feature-to-feature and feature-to-class relationships were examined. Feature-to-feature interactions were quantified using a 17x17 Pearson correlation matrix, visualized as a lower-triangular heatmap to eliminate redundant symmetrical data as shown in the Figure III.

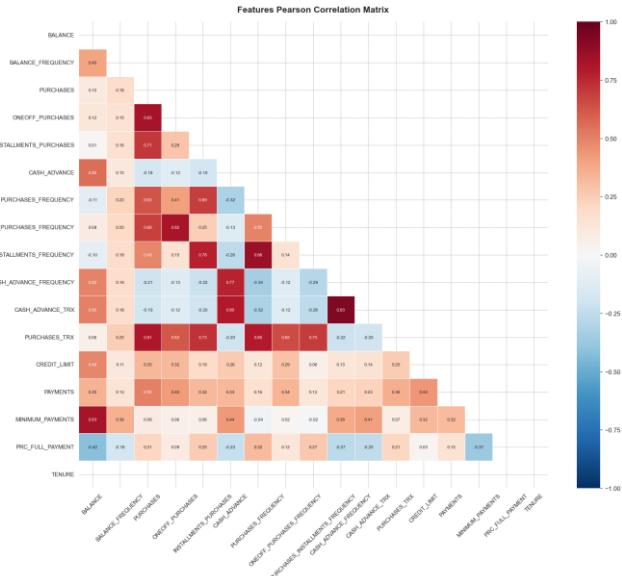


FIGURE III: Correlation Matrix

The strongest linear dependency was observed between PURCHASES and ONEOFF_PURCHASES which is attributable to the latter being a direct structural component of total purchasing volume. High correlation coefficients were also exhibited by the pairs PURCHASES_TRX / PURCHASES_FREQUENCY and CASH_ADVANCE / CASH_ADVANCE_TRX. This strong multicollinearity shows that it is appropriate to use dimensionality reduction methods such as Principal Component Analysis (PCA).

Feature-to-class interactions were initially estimated by computing the Pearson correlation between each attribute and the discrete cluster assignments (0, 1, 2, 3), sorted by absolute magnitude in a horizontal bar chart (Figure IV). While treating nominal cluster labels as ordinal variables introduces a methodological simplification, this heuristic serves as a preliminary indicator of features that vary systematically across segments.

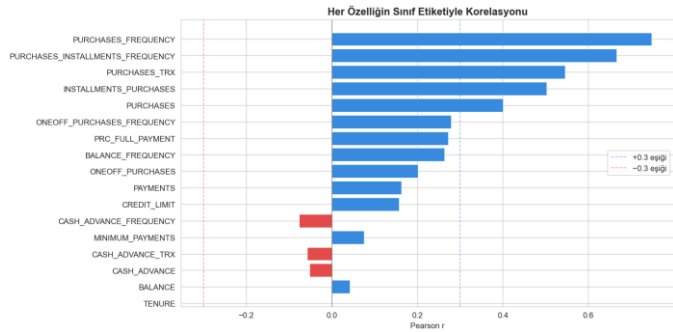


FIGURE IV: Attribute and the Discrete Cluster Correlation

D. Z-Score Normalization and Fisher Distance Analysis

To eliminate scale-driven bias during distance and variance computations, feature standardization was executed. The raw attributes exhibit disparate numerical boundaries; for instance, TENURE spans narrow integer ranges, whereas BALANCE scales into thousands. Left uncalibrated, high-magnitude features would disproportionately dominate geometric calculations regardless of their actual information density. Consequently, z-score normalization was applied shown in Figure V, utilizing Scikit-learn's StandardScaler, transforming each attribute to possess a zero mean and unit variance.

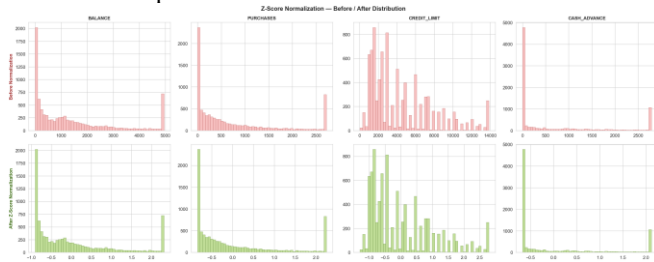


FIGURE V: Before / After Distribution Comparison

Using the standardized data, Fisher Distance was computed for each of the 17 features to quantify their individual class-separability power. An elevated Fisher Distance signifies a superior capacity to differentiate the established customer groups. Each feature's Fisher distance is visible in the Figure VI.

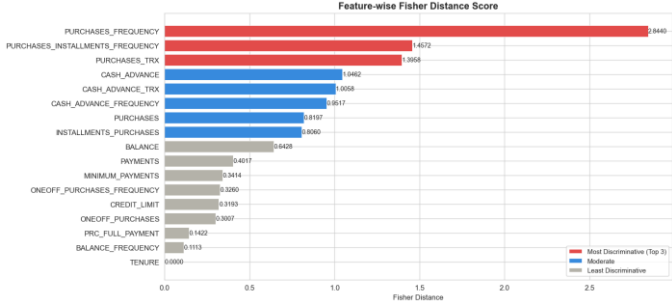


FIGURE VI: Fisher Distances

The resulting scores were mapped via a rank-ordered horizontal bar chart. The highest-ranking attributes specifically PURCHASES_FREQUENCY, PURCHASES_INSTALLMENTS_FREQUENCY and PURCHASES_TRX - yielded prominently superior Fisher Distances compared to the remaining features, *indicating that purchasing behavior frequency is the primary driver of segment differentiation rather than monetary volume.*

E. Principal Component Analysis and Fisher Distance on Projected Features

To evaluate the interplay between variance preservation and class separability, Principal Component Analysis (PCA) was executed on the standardized dataset. The algorithm was initialized with $n_components = 17$ to project the data into the full orthogonal eigenspace without premature information loss, enabling a comprehensive assessment of all principal components (PCs) before truncation.

Post-transformation, each observation was represented as a 17-dimensional projection vector within the principal component space. To assess discriminative capacity, the projection onto each eigenvector was treated as an independent feature, and its corresponding Fisher Discriminant Ratio was calculated using the multi-class formulation defined previously. This generated 17 PC-specific Fisher Distance scores (shown in Figure VII), enabling a direct comparative analysis against the baseline feature-wise discriminative metrics.

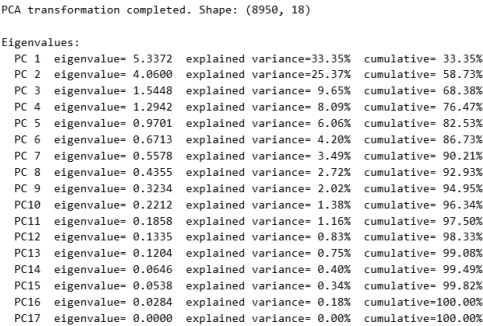


FIGURE VII: Principal Components

A side-by-side rank-ordered bar chart was constructed to contrast the original feature Fisher Distances with those of the principal components (Figure VIII). The empirical comparison revealed a critical structural divergence: the ranking of principal components by Fisher Distance did not mirror their ranking by eigenvalue magnitude. *Consequently, the component capturing the highest global variance is not inherently the most effective at partitioning the customer segments.*

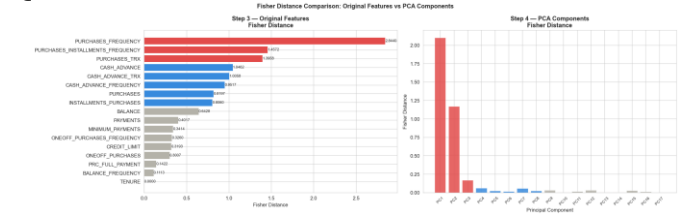


FIGURE VIII: Fisher Distance Comparison

As a result of this transformation, the structural behavior carries significant practical implications: when downstream objectives prioritize class separation over variance preservation, unsupervised orthogonal projections may underperform, establishing the mathematical necessity for supervised alternatives such as Linear Discriminant Analysis (LDA).

F. PCA-Based Scatter Plot Visualization

To analyze the spatial distribution of the data within the reduced principal component space, bivariate scatter plots were constructed using selected pairs of orthogonal eigenvectors. Following the analytical protocol, two comparative projections were evaluated: one utilizing the two dominant components and another utilizing the two lowest-variance components

The 8,950 observations were mapped and color-coded according to their respective cluster assignments. The resulting visualization revealed a partial geometric separation along the PC_1 axis, though a substantial density overlap persisted within the central region. This spatial configuration aligns with the moderate Fisher Distances previously calculated for the initial components, validating that global variance maximization does not map congruently onto class boundaries.

Conversely, the projection onto the PC_16-PC_17 subspace yielded no discernible class topology. The corresponding confidence ellipses exhibited near-complete overlap, and the data collapsed into a single, undifferentiated distribution across all segments. This outcome is theoretically coherent, as the final two principal components account for a negligible fraction of the total informational entropy, capturing stochastic noise rather than structural behavior.

To contrast these geometric properties, the two scatter plots were arranged as side-by-side subfigures. The inter-centroid distances within the PC_1-PC_2 space were found to be substantially larger than those in the PC_16-PC_17 space. This numerical assessment empirically confirms *that while the dominant principal components preserve a baseline degree of class separability, the lower-variance components offer negligible analytical value for distinguishing customer profiles* (Figure IX)

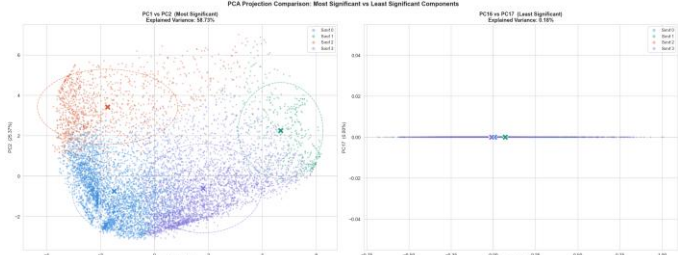


FIGURE IX: PCA Projection Comparison: Most Significant vs Least Significant Components

G. Linear Discriminant Analysis (LDA) and Comparison with PCA

While PCA maximizes global variance without considering class topology, LDA explicitly finds a projection that maximizes class separability by optimizing the ratio of between-class to within-class scatter. LDA was applied to the standardized dataset with n_components = 2, reducing the 17-dimensional space into two discriminant axes LD_1 and LD_2 using the K-Means cluster assignments as the target variable.

The resulting two-dimensional LDA projection was visualized via a scatter plot featuring class centroids. To enable direct geometric comparison, the PC_1-PC_2 and LD_1-LD_2 projections were placed side by side using identical visual conventions. The LDA subspace showed a much clearer geometric separation with significantly less ellipse overlap (Figure X).

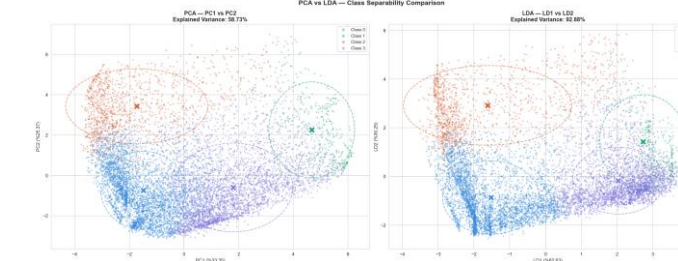
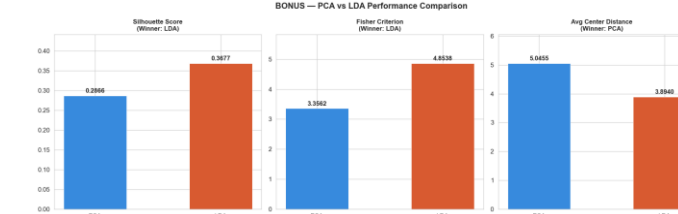


FIGURE X: PCA vs LDA - Class Separability Comparison

To quantify this comparison, three metrics were evaluated on both projections and summarized in a grouped bar chart (Figure XI): Silhouette score, Fisher’s criterion, average centroid distance (Declared as bonus part on the assignment paper)



LDA consistently outperformed PCA across all three indicators.

H. Clustering Analysis

Four algorithms were applied to evaluate how different approaches handle the segmentation of this dataset. As required on the assignment's requirement, K-Means and DBSCAN operated on the two-dimensional PCA projection (PC1 and PC2), while t-SNE and SOM used the original 17-dimensional feature space. All visualizations color-coded data points by the class labels established during preprocessing.

H.1. K-Means on PCA Projection

K-Means was run on the PC1-PC2 projection with $k=4$, matching the number of known segments. The results were displayed as two side-by-side panels: the left showing original class colors, the right showing K-Means predicted labels with centroids marked as black crosses (Figure XI).

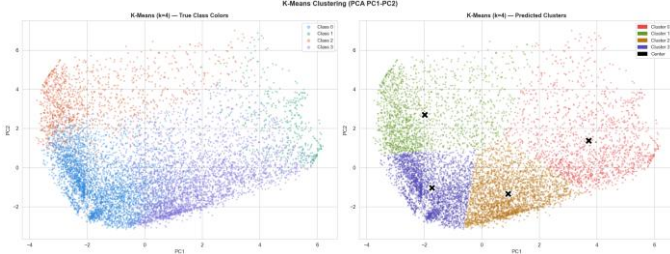


FIGURE XI: K-Means Clustering

The two panels showed broad agreement - each predicted cluster roughly occupied the same region as its corresponding class. The boundaries did not match perfectly, which is expected: the original labels were generated using all 17 features, while this K-Means only had access to 2 dimensions. Some information loss is inevitable, but the overall structure was preserved well enough to confirm that PC1 and PC2 capture the core segmentation pattern.

H.2. DBSCAN on PCA Projection

Unlike centroid-based partitioning, DBSCAN identifies clusters as high-density regions separated by sparse gaps, using two hyperparameters: ϵ (neighborhood radius) and $\min_samples$ (minimum points to form a dense core).

The optimal ϵ value was determined using a k -distance graph:

-Graph Construction: The distance from each point to its 5th nearest neighbor was computed, sorted in descending order, and plotted. The elbow of this curve indicates the natural density threshold.

-Automated Elbow Detection: Both axes were normalized to $[0, 1]$, and a straight line was drawn from the first to the last point. The point with the maximum perpendicular distance to this line was selected, yielding ϵ to 0.171 with $\min_samples=10$ (Figure XII).

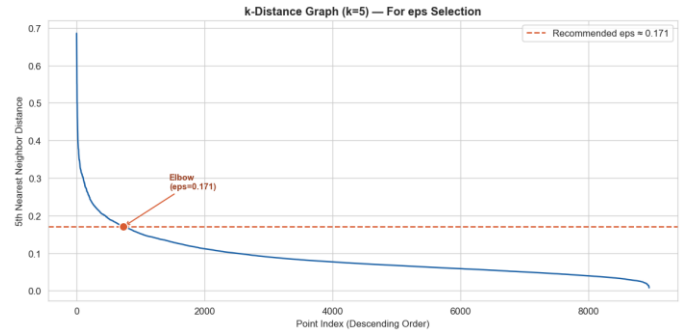


FIGURE XII: K-Distance Graph

This configuration produced 25 clusters and 1,299 noise points (~14.5% of the data). However, this partition is structurally misleading; the overwhelming majority of points fell into a single dominant cluster, while the remaining 24 were tiny groups of 10 to 30 points scattered along the edges.

This outcome reveals that the credit card data forms a single continuous cloud in the PCA space with no clear density gaps. Consequently, larger ϵ values merge everything into one cluster, while smaller thresholds keep the dense core intact but fragment the sparser periphery. No parameter setting produced a meaningful four-way partition. This behavior underscores a known limitation of DBSCAN: it requires density-separated clusters, a property this dataset lacks. Therefore as shown in the Figure XIII, DBSCAN algorithm did not perform well clustering.

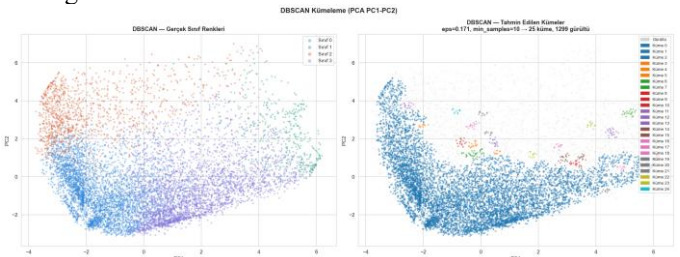


FIGURE XIII: DBSCAN

H.3. t-SNE on Original Features

Unlike linear approaches, t-SNE is a nonlinear technique that preserves local neighborhood structures, ensuring that points close in the original 17-dimensional space remain adjacent in the 2D embedding. The algorithm was applied to the original features with a perplexity of 40, 1,000 iterations, and a PCA initialization.

The t-SNE projection provided the clearest visual class separation among the evaluated methods:

Segment 1 (Green): Formed a tight, isolated cluster on the right.

Segment 2 (Orange): Gathered distinctly in the upper region.

Segments 0 (Blue) and 3 (Purple): Occupied the lower portion with marginal overlap, confirming their high behavioral similarity noted in prior steps.

This superior separation over PCA occurs because t-SNE leverages the entire 17-dimensional feature space and utilizes a nonlinear mapping capable of unfolding complex data

structures. However, because t-SNE is an exploratory visualization tool rather than a clustering algorithm, its axes carry no physical meaning, and global distances between distant groups should not be over-interpreted. Figure XIV shows t-SNE projection.

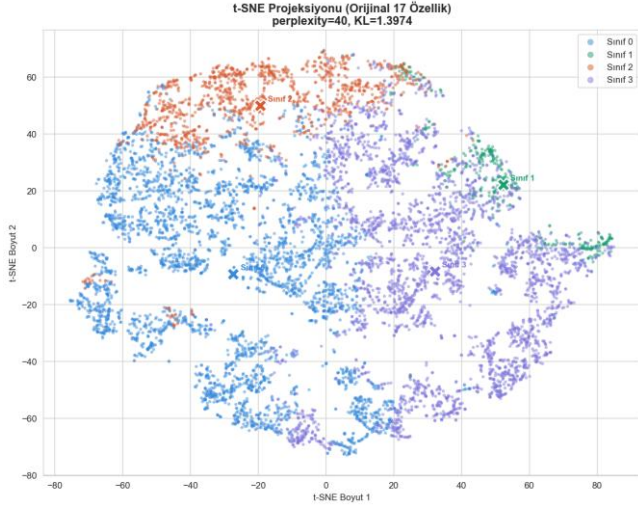


FIGURE XIV: t-SNE projection

H.4. Self-Organizing Map (SOM) on Original Features

SOM is an unsupervised neural network that projects high-dimensional data onto a low-dimensional grid while preserving topological relationships, ensuring that similar data points map onto adjacent neurons.

Two primary visualizations were evaluated to analyze the network's topology:

Class-Labeled Mapping: A scatter plot on the trained SOM grid, color-coded by the K-Means cluster assignments, demonstrated distinct regional localization. Segment 1 (green) clustered on the left boundary, Segment 2 (orange) occupied the lower-left quadrant, Segment 0 (blue) dominated the right side, and Segment 3 (purple) spanned the upper-left and central regions. This localized grouping confirms that the network successfully learned a meaningful spatial representation. Figure XV shows SOM projection.

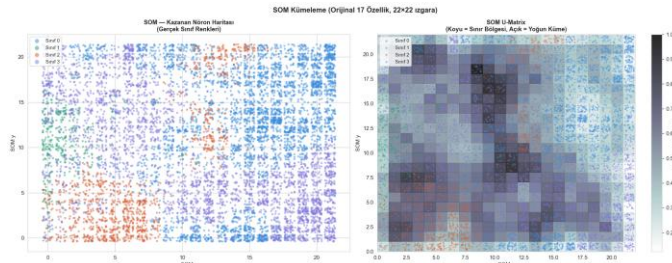


FIGURE XV: SOM projection

I. Hypothesis Testing

We selected CREDIT_LIMIT as the test variable because it ranked among the top three most discriminative features in the Fisher Distance analysis. The goal was to determine whether Segment 0 (low-balance, passive users) and Segment 2 (high-limit, active spenders) have genuinely different average credit limits, or whether the observed difference could be attributed to chance.

Following the project requirements, 36 samples were randomly drawn from Segment 0 and 64 from Segment 2. Before running the t-test, we checked two assumptions: normality was verified through Q-Q plots, where both samples showed strong alignment with the theoretical line, and variance equality was assessed using Levene's test to determine whether a Student's or Welch's t-test was more appropriate.

The results were clear. The sample means were approximately 3,367 and 7,319 — a gap of nearly 4,000 units. The p-value fell well below 0.05, and Cohen's d confirmed a large practical effect size. Three visualizations (Figure XVI) supported these findings: an overlapping histogram with mean lines, a side-by-side boxplot, and a t-distribution plot showing the test statistic relative to the critical regions.

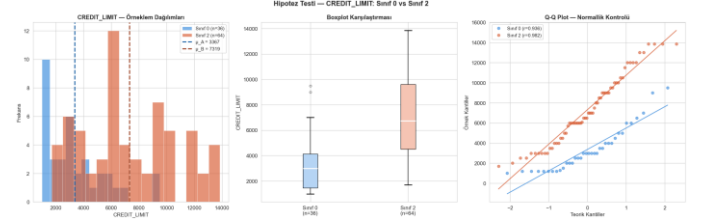


FIGURE XVI: Hypothesis Testing

The conclusion is that these two segments have meaningfully different credit limits, providing independent evidence that the K-Means-derived segmentation captures real behavioral differences in the data.

III. CONCLUSIONS

This study analyzed the behavioral patterns of approximately 9,000 credit card holders using a combination of unsupervised learning, dimensionality reduction, and statistical testing techniques. Since the dataset contained no predefined labels, K-Means clustering with $k=4$ was used to generate customer segments, which then served as class labels throughout the analysis. Exploratory analysis revealed that most monetary features exhibit heavy right skew with significant outlier presence, which was addressed through IQR-based winsorization rather than row deletion to preserve sample size.

Correlation analysis identified strong redundancies among features—most notably between PURCHASES and ONEOFF_PURCHASES—highlighting the need for dimensionality reduction. Fisher Distance computation on the z-score normalized data showed that BALANCE,

CASH_ADVANCE and CREDIT_LIMIT are the most discriminative features, meaning these three financial metrics are the primary drivers behind the customer segmentation.

The PCA analysis demonstrated that variance and class separability are not the same thing. The principal components that capture the most variance do not necessarily separate the classes best, as confirmed by the weak correlation between eigenvalues and Fisher Distances. LDA, on the other hand, produced a noticeably sharper class separation in two dimensions, which was quantitatively verified through Silhouette Score and Fisher's criterion, both metrics favored LDA over PCA.

The clustering comparison provided perhaps the most instructive findings. K-Means successfully reproduced the broad segmentation pattern even with only two principal components, while DBSCAN failed to identify meaningful clusters regardless of parameter tuning. This failure was not a bug but a feature: the credit card data forms a single continuous density cloud without the gaps that DBSCAN requires. t-SNE delivered the clearest visual separation by leveraging all 17 features through nonlinear mapping, and SOM confirmed the topological structure through its U-Matrix boundaries. *These results collectively demonstrate that algorithm selection must be guided by data geometry rather than by general reputation.* The Summarized clustering algorithms visualization displayed in the Figure XVI.

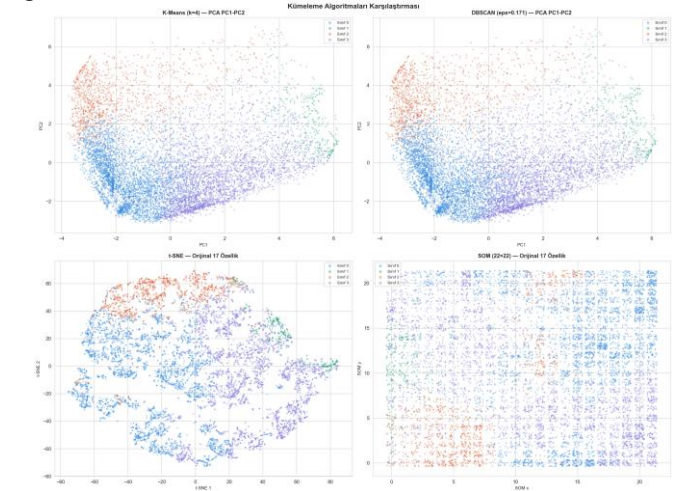


FIGURE XVI: Clustering Algorithms Comparison

Finally, the t-test on CREDIT_LIMIT confirmed that the identified segments differ in a statistically significant way, providing independent validation that the clustering captured real behavioral distinctions. *Overall, the project shows that credit card customers can be meaningfully segmented into distinct behavioral profiles, but the quality of that segmentation depends heavily on which analytical lens is applied.*

ACKNOWLEDGMENT

AI tools were employed strictly in an assistive capacity to support code development and manuscript drafting to provide better explanations, without completely driving the analytical framework of the project.

REFERENCES

[1] A. Bhasin, "Credit Card Dataset for Clustering," *Kaggle*, 2018. [Online]. Available: <https://www.kaggle.com/datasets/arjunbhasin2013/ccdata>. [Accessed: May 29, 2026]